<u>Thm</u> Let (X, d) be a metric space. If X is compact, then every sequence in X has a subsequence that converges in X.

<u>Pf</u>: Suppose not: X is compact, and there exists a sequence $\{x_n\}_{n \geq 1}$ s.t. every subsequence does not converge.

Last time:

① $\{x_n : n \in \mathbb{N}\}$ is infinite

② $\forall x \in X$, $\exists r_x > 0$ s.t. $B_{r_x}(x)$ does not intersect $\{x_n : n \in \mathbb{N}\} \setminus \{x\}$.

Then $\{B_{r_x}(x)\}_{x \in X}$ is an open cover for X. As X is compact, $\exists$ a finite subcover

$$X \subseteq B_{r_{y_1}}(y_1) \cup \ldots \cup B_{r_{y_N}}(y_N).$$

But each $B_{r_x}(x)$ contains at most one element of $\{x_n : n \in \mathbb{N}\}$. This contradicts that the set $\{x_n : n \in \mathbb{N}\}$ is infinite. $\square$

<u>Cor</u> Let (X, d) be a metric space. If $A \subseteq X$ is compact, then A is closed and bounded.

<u>Def</u> We say $A \subseteq X$ is <u>bounded</u> if $\exists M > 0$ s.t. $d(x, y) \leq M$ $\forall x, y \in A$.

- I.e., the set $\{d(x, y) : x, y \in A\}$ of real numbers is bounded.

<u>Pf</u>: Closed: Fix $a \in \bar{A}$. Then $\exists$ a sequence $\{a_n\}_{n\geq 1}$ in A that converges to a. By the theorem, $\exists$ a subsequence $\{a_{k_n}\}_{n\geq 1}$ that converges to some point $\tilde{a} \in A$. Together, we conclude $a = \tilde{a} \in A$. So $\bar{A} \subseteq A$, and thus A is closed.

Bounded: Suppose not — A is unbounded. Fix $x_1 \in A$. Note that $\{d(x_1, y) : y \in A\}$ is not bounded above, since otherwise

$$d(x_1, y) \leq L \quad \forall y \in A \quad \Rightarrow \quad d(x, y) \leq d(x, x_1) + d(x_1, y) \leq 2L \quad \forall x, y \in A$$

$$\Rightarrow A \text{ is bounded}$$

So, $\forall L > 0 \ \exists y \in A$ s.t. $d(x_1, y) > L$. We construct a sequence $\{x_n\}_{n\geq 1}$ as follows:

- Take $L = 1 \Rightarrow \exists x_2 \in A$ s.t. $d(x_1, x_2) > 1$
- Take $L = 1 + d(x_1, x_2) \Rightarrow \exists x_3 \in A$ s.t. $d(x_1, x_3) > 1 + d(x_1, x_2)$

Proceeding inductively, we get a sequence $\{x_n\}_{n\geq 1}$ s.t.

$$d(x_1, x_n) > d(x_1, x_{n-1}) + 1 \quad \forall n \geq 1$$

$$\begin{aligned} \Rightarrow \quad d(x_1, x_n) &> d(x_1, x_{n-1}) + 1 \\ &> d(x_1, x_{n-2}) + 1 + \underbrace{1}_{>0} \\ &\ \ \vdots \\ &> d(x_1, x_m) + 1 \quad \forall n > m \end{aligned}$$

$$\Rightarrow d(x_1, x_n) \leq d(x_1, x_m) + d(x_m, x_n)$$
$$< d(x_1, x_n) - 1 + d(x_m, x_n)$$

$$\Rightarrow d(x_m, x_n) > 1 \quad \forall n > m$$

So no subsequence of $\{x_n\}_{n \geq 1}$ can be Cauchy. This contradicts the previous theorem. $\square$

<u>Next</u>: $X = \mathbb{R}^n$

<u>Thm</u> Let $n \in \mathbb{N}$. For any $1 \leq p, q \leq \infty$, the metrics d_p and d_q are <u>equivalent</u>: $\forall A \subseteq \mathbb{R}^n$,

$$A \text{ open in } (\mathbb{R}^n, d_p) \iff A \text{ open in } (\mathbb{R}^n, d_q).$$

- HW8 #2: proof for $n = 2$, $p = 1$, $q = \infty$.
- HA8: full proof
- So, in the following, it doesn't matter which $1 \leq p \leq \infty$ we pick
- Goal:

<u>Thm</u> Every set of the form $[a_1, b_1] \times \ldots \times [a_n, b_n]$ is compact in $\mathbb{R}^n$ (with the metric d_p for any $1 \leq p \leq \infty$).

- Before we start the proof, here are some important consequences:

<u>Thm</u> (Heine-Borel) For any $A \subseteq \mathbb{R}^n$,

$$A \text{ is compact} \iff A \text{ is closed and bounded.}$$

<u>Pf</u>: $\Rightarrow$|: Previous Cor. (True for <u>any</u> metric space.)

$\Leftarrow$|: Suppose A is closed and bounded.

Then $\exists M > 0$ s.t. $A \subseteq [-M, M] \times \ldots \times [-M, M] = [-M, M]^n$.

We know $[-M, M]^n$ is compact by the previous theorem. As A is a closed subset of $[-M, M]^n$, then A is compact by HW10. $\square$

<u>Cor</u> Let $A \subseteq \mathbb{R}^n$. The following are equivalent:

① A is compact.

② Every sequence in A has a subsequence that converges in A.

③ A is closed and bounded.

<u>Pf</u>:

$$\begin{tikzcd} & ② \arrow[dr, Rightarrow, "\text{Last time}"] & \\ ① & & ③ \arrow[ll, Rightarrow, "\text{Heine-Borel}"] \arrow[ul, Rightarrow, "\text{Last time}"'] \end{tikzcd}$$

$\square$

<u>Rmk</u> For an arbitrary metric space (X, d),

$$① \overset{\text{always true}}{\Longleftrightarrow} ② \underset{\text{sometimes false!}}{\overset{\Rightarrow}{\not\Leftarrow}} ③$$

E.g., $A = \{x \in \mathbb{Q} : 0 \leq x \leq 1\}$ is...

- closed and bounded in $\mathbb{Q}$:

$$\bar{A}^{\mathbb{Q}} = \bar{A}^{\mathbb{R}} \cap \mathbb{Q} = [0, 1] \cap \mathbb{Q} = A$$

- not compact in $\mathbb{Q}$:

$$\begin{aligned} A \text{ compact in } \mathbb{Q} &\Rightarrow A \text{ compact in } \mathbb{R} \\ &\Rightarrow A \text{ closed in } \mathbb{R} \\ &\Rightarrow A = \bar{A}^{\mathbb{R}} = [0,1] \neq A, \end{aligned}$$

a contradiction.

- Next: $[a_1, b_1] \times \ldots \times [a_n, b_n]$ is compact in $\mathbb{R}^n$
- First step:

<u>Prop</u> If $[a_k, b_k] \subseteq \mathbb{R}$ are intervals s.t.

$$[a_k, b_k] \supseteq [a_{k+1}, b_{k+1}] \quad \forall k \in \mathbb{N},$$

then $\bigcap_{k \in \mathbb{N}} [a_k, b_k] \neq \emptyset$.

<u>Pf</u>: Note that

$$a_1 \leq a_2 \leq \ldots \leq a_k \leq b_k \leq \ldots \leq b_2 \leq b_1 \quad \forall k \geq 1.$$

So the set $\{a_k : k \in \mathbb{N}\}$ is...

- nonempty: it contains a_1
- bounded above: by b_1

Therefore $x = \sup\{a_k : k \in \mathbb{N}\}$ exists in $\mathbb{R}$.

Claim: $x \in \bigcap_{k \in \mathbb{N}} [a_k, b_k]$. Note that

- $x \geq a_k$: since x is an upper bound for $\{a_k : k \in \mathbb{N}\}$
- $x \leq b_k$: since b_k is an upper bound for $\{a_k : k \in \mathbb{N}\}$

So $x \in [a_k, b_k]$ $\forall k$. $\square$